

Seasoned Data Scientist and Machine Learning Engineer with 20+ publications who turns large, messy real-world data into deployed decisions. Track record spans the full lifecycle: feature engineering on billion-row datasets, survival and causal modeling, deep learning, NLP, and production drift monitoring. All built on an engineering foundation.

Experience

- **Senior Machine Learning Engineer**
Visual Thought Labs

 - Architected and built secure distributed containerized edge-prediction network of auto-connecting field devices. Used WireGuard encrypted mesh with ZeroMQ/MQTT messaging, and deployed across 3 Nvidia Jetson nodes.
 - Evaluated a vision-language model (VLM) for syringe inference; with prohibitive edge-hardware cost, pivoted the VLM to run offline on logged data as an automatic labeler — scaling training-data annotation by ~3.5x.

October 2025 - Present
San Francisco, CA
- **Senior Data Scientist**
UCSF

 - Built 6-hospital data-drift monitor adopted by ~80 researchers, wired into a CI/CD pipeline that auto-triggers retraining.
 - Directed a matched-cohort study across UC Health, Northwestern, and Harvard, published in JAMA Network Open on traumatic-brain-injury outcomes, quantifying a 67% higher risk of malignant brain tumors (HR=1.67).

January 2022 - October 2025
San Francisco, CA
- **Machine Learning Engineer**
Syntegra

 - Created a GenAI synthetic patient data generator that transformed structured EHR records into natural-language sentences, trained BERT models, and regenerated structured data, enabling safe data sharing for rare-case research.

October 2021 - January 2022
San Francisco, CA
- **Lead Data Scientist**
LifeBell AI

 - Led a 5-person data science team building production clinical-risk models, owning technical direction, code review, and delivery. Deployed at Naval Medical Center San Diego yielding sepsis response ~2 hours earlier.
 - Built a bespoke training-to-registry pipeline (dataset query → preparation → training → model export → validation scoring → versioned artifacts) and the Docker/AWS deployment that ships from it.

December 2020 - October 2021
Atlanta, GA (remote)
- **Data Scientist**
UCSF

 - Led feasibility analyses that helped secure \$5M+ in research partnerships (J&J, Roche, Genentech).
 - Designed and shipped PySpark clinical text de-identification pipeline to process 600M+ notes used by 400+ researchers.

August 2018 - December 2020
San Francisco, CA
- **Embedded Systems Engineer**
NASA Education & Public Outreach

 - Conceived/built/launched EdgeCube 10cm satellite, overseeing sensor integration, data logging, and radio telemetry.

May 2014 - August 2018
Rohnert Park, CA

Education

- **Stanford University** MS in Computational and Mathematical Engineering 2018
- **Sonoma State University** BS in Physics and Mathematics 2014

Skills

- **Languages** Python, SQL, Julia, C, C++, JavaScript, TypeScript, $\text{\LaTeX}$, Bash
- **ML Frameworks** PyTorch, TensorFlow, Scikit-Learn, XGBoost, JAX, Ollama, HuggingFace, StatsModels
- **Infrastructure/Compute** Spark, Spark-SQL, Hadoop, MPI, OpenMP, CUDA, AWS, GCP, Docker
- **Visualization** Dash, Plotly, Matplotlib, Seaborn

Publications (Selected)

- C. Halabi, **H. Mills**, A. M. DiGiorgio, et al. “Civilian Traumatic Brain Injury Is Associated with Longitudinally Increased Risk of PTSD, Suicidality, and Other Mental Health Diagnoses”. In: *Journal of Neurotrauma* (2026), p. 08977151251414145.
- A. S. Tang, K. P. Rankin, G. Ceroni, S. Miramontes, **H. Mills**, J. Roger, et al. “Leveraging electronic health records and knowledge networks for Alzheimer’s disease prediction and sex-specific biological insights”. In: *Nature Aging* 4.3 (2024), pp. 379–395.
- M. A. Aras, S. Abreau, **H. Mills**, L. Radhakrishnan, et al. “Electrocardiogram detection of pulmonary hypertension using deep learning”. In: *Journal of Cardiac Failure* 29.7 (2023), pp. 1017–1028.